

msaGUI User Guide

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QUICK START GUIDE

INSTALLATION

Download the msaGUI folder directly from <https://github.com/cdavislab/multispectral-analysis> via Code > Download ZIP. Once downloaded, unzip the folder. If your operating system or version is not supported, follow the instructions under Advanced Installation.

RUNNING msaGUI

Open msaGUI using the executable for your operating system (e.g. "msaGUI_Windows" or "msaGUI_Mac")

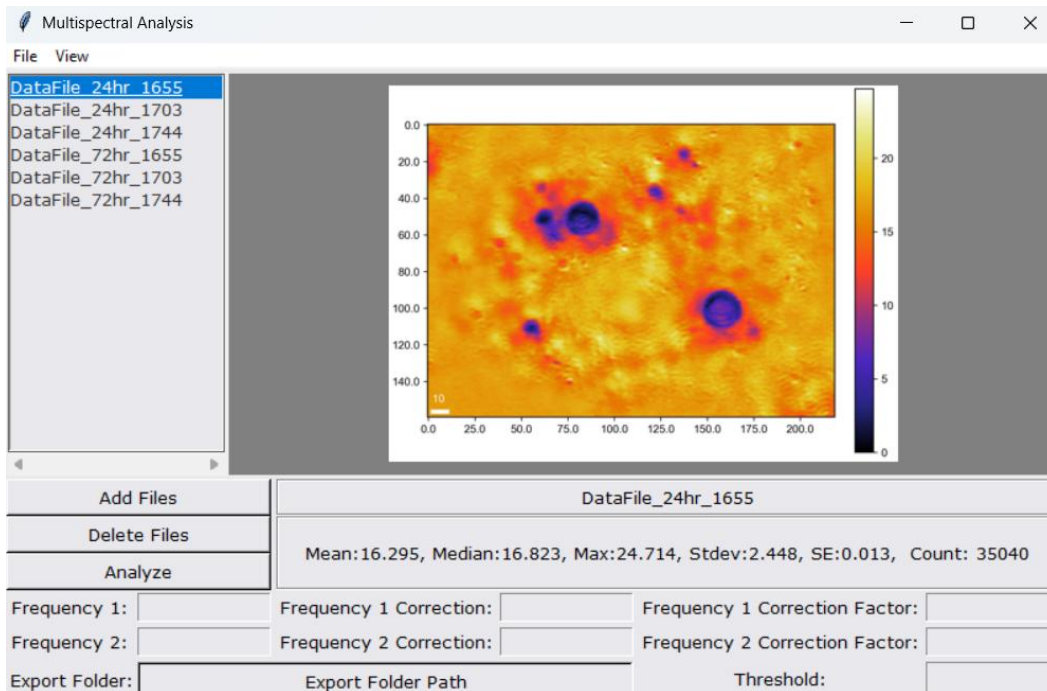
LOADING DATA

To visualize and prepare data for ratioing, images need to be initially loaded into the program.

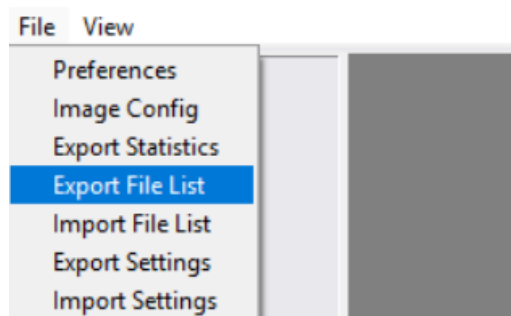
1. Click "Add Files"
2. Select one or multiple comma delimited files. For the test data, select the six data files available.
3. If files are not visible, ensure that **View > Show Single-Wavenumber** is checked. To view just the file names rather than full paths, select **View > File Path > View File Only**.

VIEWING RAW DATA

Once data files are loaded, click on the filenames to view the images. To view histograms instead, go to **View > Histogram** or press Ctrl + H. At this point, the GUI should look like this:



EXPORTING FILE LIST



Export the file list for later viewing via **File > Export File List**.

ANALYZING DATA

The program will automatically group files together by similar filenames after excluding any labels put in the Frequency 1, Frequency 1 Correction, Frequency 2, or Frequency 2 Correction. See below for example:

Frequency 1:	1703	Frequency 1 Correction:	1655	Frequency 1 Correction Factor:	0.34
Frequency 2:	1744	Frequency 2 Correction:	1655	Frequency 2 Correction Factor:	0.66
Export Folder:	Export Folder Path			Threshold:	0.10

The example data are labeled by the wavenumber the image was taken at. E.g. "DataFile_24hr_1655" indicates that the image was taken at 1655 cm^{-1} . With the settings below, the intensities of the 1703 cm^{-1} and 1744 cm^{-1} images are subtracted by 34% and 66% of the 1655 cm^{-1} image intensity respectively. Afterward, the corrected images of 1703 cm^{-1} and 1744 cm^{-1} are ratioed. In mathematical notation, where the I is the output intensity:

$$I_{ratio} = \frac{I_{1703} - 0.34 \cdot I_{1655}}{I_{1744} - 0.66 \cdot I_{1655}}$$

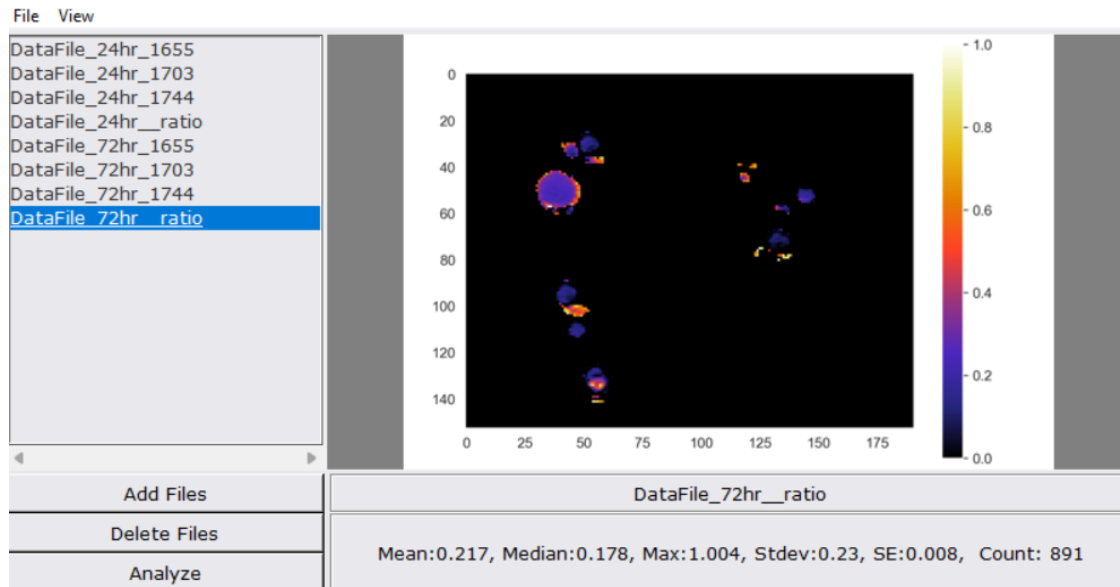
The image file names must be the same, excluding keywords. In this example, DataFile_24hr_1744, DataFile_24hr_1703, DataFile_24hr_1655 all share a common name except for the keywords specific by the user-defined parameters.

In this example, the threshold is set to .10. Thus, any values in the 1744 cm^{-1} image that are less than 10% of the maximum value found in the image will be automatically set to 0 in the final ratio image.

To analyze data,

1. Configure labels, correction factors, and threshold as demonstrated above.
2. Select all files in the file list. To do so, click on one file in the list and press Ctrl + A. Alternatively, left click the top file and then, while holding down Shift, left click the bottom file.
3. Verify that all six example data files are selected (they should be highlighted)
4. Click "Analyze" and wait for the loading bar to reach 100%.
5. After analysis, the ratio images appear in the file list. If they are not visible, ensure that **View > Show Ratios** is checked.

VIEWING ANALYZED DATA AND EXPORTING DATA



Now that the files are analyzed, the output images are viewable. To see all images from the same group in a single window, change the view to group view by clicking **View > Group View**.

EXPORTING FILES AND STATISTICS

Files are automatically saved on the computer in the subfolder "msa_analysis" under the folder where code is running. To change the export folder, click on the bottom left button in the GUI.

Frequency 1:		Frequency 1 Correction:		Frequency 1 Correction Factor:	
Frequency 2:		Frequency 2 Correction:		Frequency 2 Correction Factor:	
Export Folder:	Export Folder Path			Threshold:	

If you change the export folder after analyzing files, a prompt will appear to ask if you want to move any previously analyzed files to the new export folder.

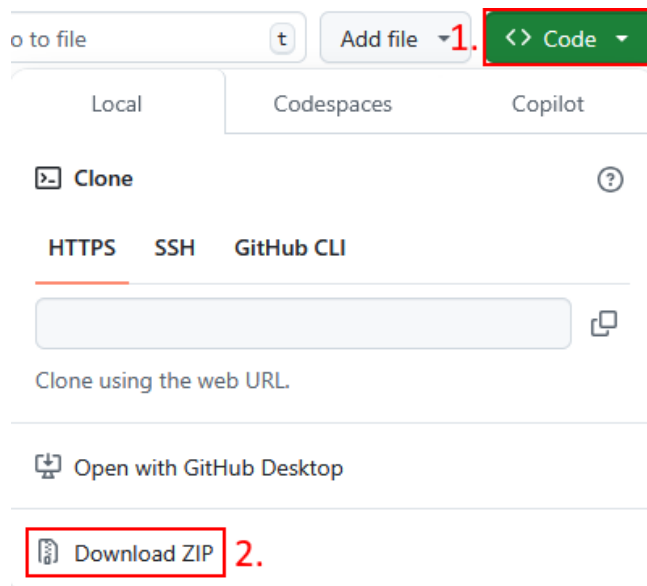
To export the statistics from each analyzed file into a spreadsheet, click **File > Export Statistics**.

INSTALLATION

ADVANCED INSTALLATION

If your operating system or version is not supported, the code can be run via its source Python code.

1. Install Python3 from <https://www.python.org/downloads/>. The code was tested primarily with Python 3.11.5.
2. Download or clone the code from <https://github.com/cdavislab/multispectral-analysis>.



3. Install the requisite packages as listed on the project's Github in requirements.txt. To do so, open up Command Prompt on Windows or Terminal on Mac and Linux. Then, type "cd FOLDER" where FOLDER is the full path to the downloaded folder (e.g. "cd C:\Users\Name\Downloads\msa_gui"). Lastly, install the requirements with the command: "pip install -r requirements.txt".
4. Run the code by typing "python msa.py" and pressing enter.

INPUTS

Input files must contain values of each pixel with each column and row pair representing a pixel's value. E.g. if a 30 x 50 pixel image should be represented with 30 columns and 50 rows.

```

1 10.97518,10.70196,10.8024,10.85626,10.56708,10.36459,10.30516
2 12.06819,14.11398,16.84036,19.26426,19.65212,20.58116,21.75483
3 12.93235,15.14095,17.89588,20.13307,20.6248,21.89983,22.77097
4 11.53448,14.8626,17.36334,19.77483,21.75422,22.34023,22.53391
5 11.09152,14.16566,16.90567,19.39143,21.96467,22.20977,22.51039
6 11.41531,14.33051,17.35577,19.48473,21.53819,23.51807,23.36234
7 11.3909,15.01118,17.96006,19.78769,21.30638,22.54708,22.44309

```

Example of how raw data is stored.

ADD/DELETE DATA

To add data, click on "Add Files" button. To delete data, click on the filename and click "Delete Files".

KEYWORDS

Frequency 1:	1703	Frequency 1 Correction:	1655	Frequency 1 Correction Factor:	0.34
Frequency 2:	1744	Frequency 2 Correction:	1655	Frequency 2 Correction Factor:	0.66
Export Folder:	Export Folder Path			Threshold:	0.10

Keyword parameters highlighted in red.

The keywords tell the program which file belong to which class (e.g. "DataFile_1703" would be a part of the Frequency 1 class in the example above). The corresponding correction factors c_f denote the proportion of the correction intensity to subtract from the primary signal. Corrections are calculated like so:

$$I = I_0 - c_f C$$

where I is the output intensity, I_0 is the original signal, and C is the correction intensity. To correct both Frequency 1 and Frequency 2 with the same correction image, input the same keyword (e.g. 1655). If a correction is not needed, input the corresponding correction factor as 0. In such a case, inputting the keyword is not necessary.

THRESHOLD

Frequency 1:	1703	Frequency 1 Correction:	1655	Frequency 1 Correction Factor:	0.34
Frequency 2:	1744	Frequency 2 Correction:	1655	Frequency 2 Correction Factor:	0.66
Export Folder:	Export Folder Path			Threshold:	0.10

Threshold parameter.

The threshold sets values of Frequency 2 that are less than that proportion of the maximum intensity to 0. These thresholded pixels are not included in statistics and they are set to 0 in the output ratio images. The inputted threshold is a proportion of the maximum intensity of Frequency 2. For example, a threshold of 0.10 will threshold all values that are less than 10% of the maximum value of Frequency 2. If that maximum value is 200, all in the Frequency 2 image that are 20 and below will be thresholded out.

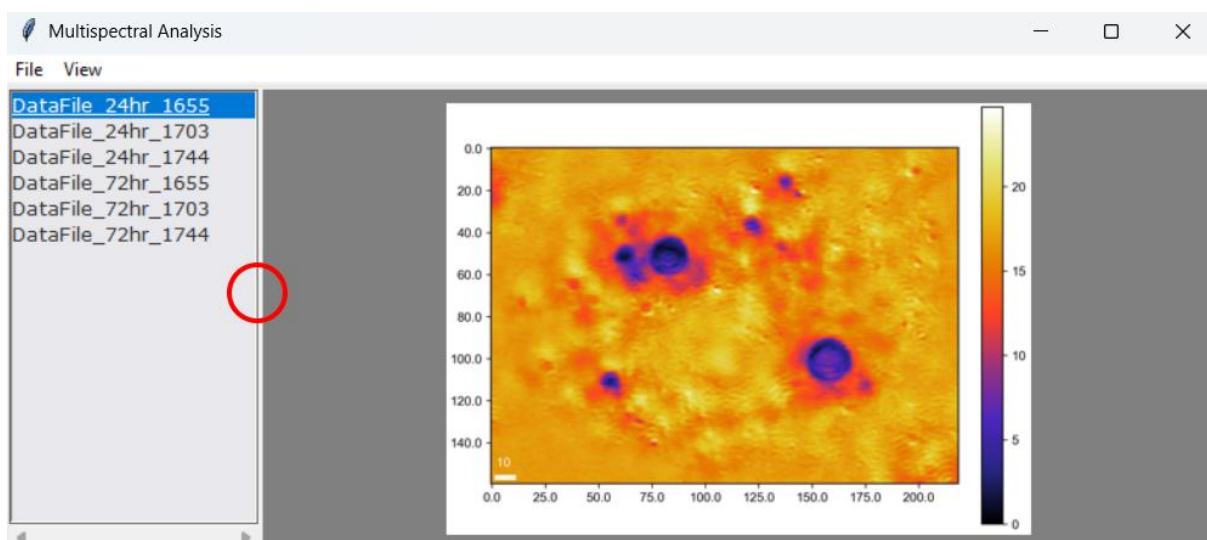
OUTPUTS

EXPORTING IMAGES AND STATISTICS

Images are automatically exported during analysis to the designated export folder. To export statistics into a single spreadsheet, click on **File > Export Statistics**. To change the export folder, click on the bottom left button of the guide as shown in the quick start guide.

VIEWING IMAGES

To view images, click on the filename in the left panel. For viewing options, select the file menu **View**. It is possible to view images by group and to view histograms. You can also choose what type of files are displayed in the file list (i.e. single-wavenumber and ratio). To adjust the image size, drag the middle of the two panes or change the window size.



Red circle marks part of divider. Drag left or right to change relative size of file list and image

FILE LIST OPTIONS

To view either the full path, just the file name, or the file name and immediate parent folder, click **View > File Path**. There are three view options: Full Path, Parent, and File Only. For the file path "C:\Users\Name\Data\DataFile_1", the different views are:

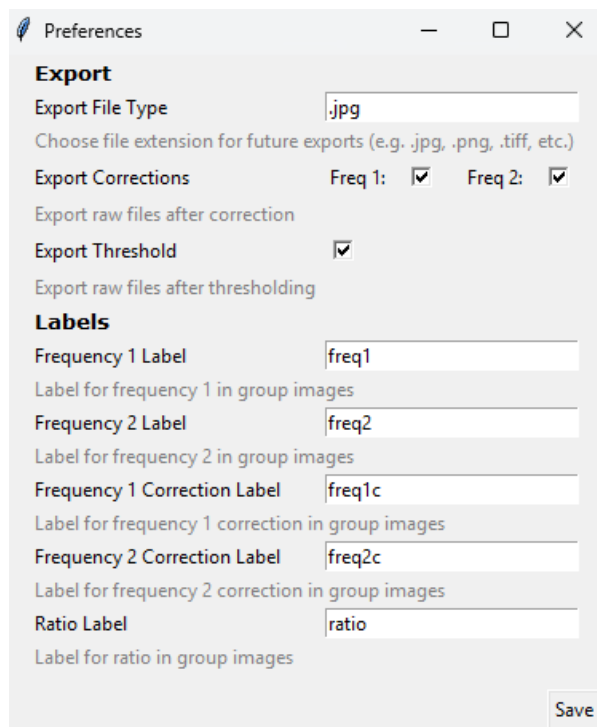
- Full Path: "C:\Users\Name\Data\DataFile_1"
- Parent: "Data\DataFile_1"
- File Only: "DataFile_1"

ADDITIONAL FEATURES

IMPORTING/EXPORTING FILE LISTS AND SETTINGS

To save file lists for later, go to **File > Export File List**. Import them with **File > Import File List**. To export Settings such as Frequency, Correction Factor, Export Folder, or Threshold, go to **File > Export Settings**. To import them, go to **File > Import Settings**. To set the defaults for program, save the file as *msa_options.txt*.

PREFERENCES



Preferences Menu

CHANGING IMAGE FILE TYPES

To change the file type that images are saved in, go to **File > Preferences**. After typing in a new file type, click Save. Changes are not saved if the window is closed using the X in the corner.

EXPORTING INTERMEDIARY FILES

In times, it may be helpful to see the intermediate steps of processing such as the images after correction or applying a threshold. The checkboxes in the preferences menu toggle whether intermediate images are saved. If checked, the images will be saved in the

same format as the input images (.csv) into the designated export folder. It is possible to reupload these into the program for easy viewing.

GROUP LABELS

It is possible to add custom labels to the group images. Leave blank for no labels.

IMAGE PREFERENCES

A multitude of aspects in outputted images are customizable in order to yield publication-ready figures. Included customizable features are below.

FONT

The current options surrounding text are the Font, Font Size, and Font Weight.

COLOR BAR

The color bar has three components: the color map, the scale, and the units.

1. Color Map: The palette used to indicate values. See a full list of available color maps here:
<https://matplotlib.org/stable/users/explain/colors/colormaps.html>.
2. Scale: The minimum and maximum values for non-ratioed images. Ratio images are automatically scaled to the minimum and maximum of the data.
3. Units: The label put next to the color bar to indicate the axis. Can be units (e.g. "mV") or full label (e.g. "Intensity (mV)").

SCALE

1. Pixel Scale: Conversion between pixels and physical unit of measurement. Setting the pixel scale calibrates the scale bar fixed value (see below).
2. Scale Bar Units: Label of units
3. Scale Bar Color: Color of scale bar and label. See a full list of available colors:
https://matplotlib.org/stable/gallery/color/named_colors.html.
4. Scale Bar Location: Position on image where the scale bar is. Valid entries are combinations of one item from each of the following lists: (upper, center, lower) and (left, center, right). For example, "upper left" or "lower center". To put a scalebar in the center, simply write "center" (as opposed to "center center").

5. Scale Bar Fixed Value: The length (in physical units) of the scalebar. For no scale bar, set this value to 0.

EXTRA

1. Number of Tick Marks: Number of tick marks on both the x and y axes. For no tick marks, set to 0.

Image Preferences

Font

Font: arial

Font of axes text

Font Size: 10

Font size of axes text

Font Weight: normal

(e.g. light, normal, heavy, bold)

Color Bar

Color Map: CMRmap

Choose Matplotlib colormap

Scale: Min 0 Max

Minimum and max of the color bar

Units:

Units of the color bar

Scale

Pixel Scale: 1

Change the scale of the pixels

Scale Bar Units:

Units of the scale bar

Scale Bar Color: white

(e.g. white or black)

Scale Bar Location: lower left

(e.g. upper/lower left)

Scale Bar Fixed Value: 10

(e.g. '10' for 10 units)

Extra

Number of Tick Marks: auto

Number of axis tick markers

Save

Image Preferences Menu